# Puma Marketing Strategies and Parkruns

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## Overview

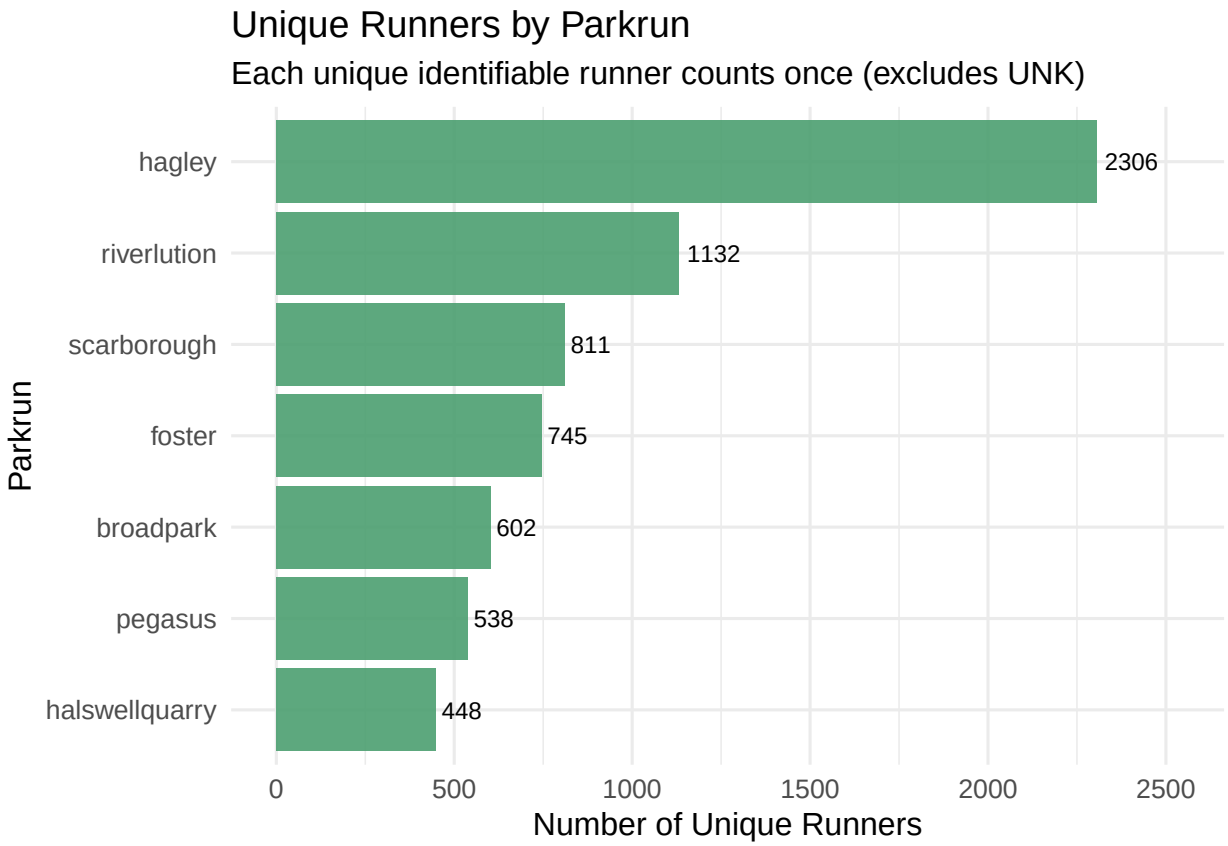
In this study, we analysed 11 weeks of Parkrun events across 7 locations in and around Christchurch/Otautahi. This study was commissioned by Puma Inc. for the purpose of determining marketing potential in these Parkruns, and developing and optimal marketing strategy. This study was conducted by Bad Ass Analytics Ltd. and this document investiagates these Parkruns in relation to the potential for marketing/advertising.

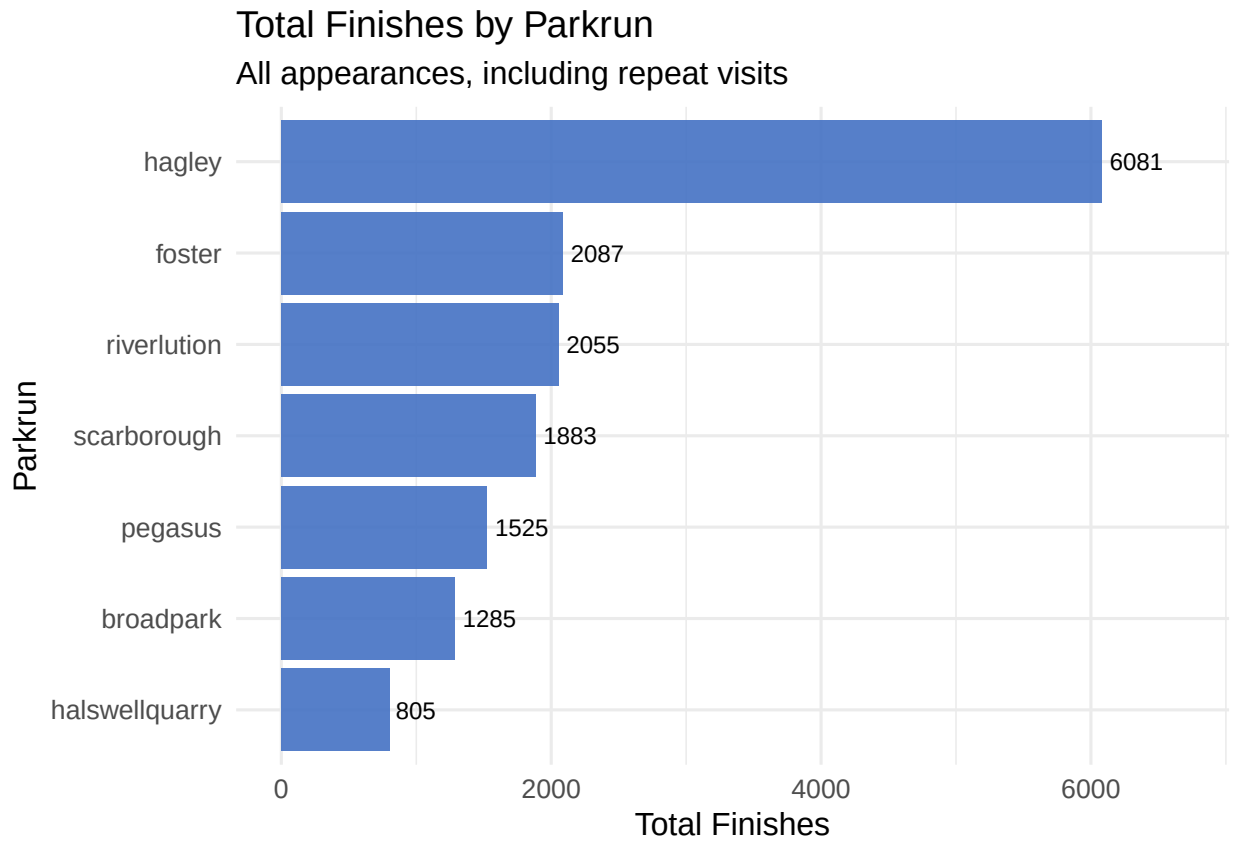
## Key Insights

### Question One: Which Parkrun provides the best marketing opportunity with the largest number of people?

Table 1: Unique runners and total finishes per Parkrun

| Parkrun | Total Finishes | Unique Runners | Unknown Finishes |
|---|---|---|---|
| hagley | 6081 | 2306 | 835 |
| riverlution | 2055 | 1132 | 162 |
| scarborough | 1883 | 811 | 218 |
| foster | 2087 | 745 | 74 |
| broadpark | 1285 | 602 | 67 |
| pegasus | 1525 | 538 | 29 |
| halswellquarry | 805 | 448 | 40 |


Unique Runners by Parkrun
Each unique identifiable runner counts once (excludes UNK)
Parkrun
hagley
riverlution
scarborough
foster
broadpark
pegasus
halswellquarry
2306
1132
811
745
602
538
448
0
500
1000
1500
2000
2500
Number of Unique Runners




```
## \nBest marketing opportunity: hagley (2306 unique runners)\n
```

### Question Two: Do new people come each week or is it always the same people?

```
## [1] "Q2: Weekly new vs returning runners (global first definition)"
```

Table 2: Weekly new vs returning runners across every Parkrun

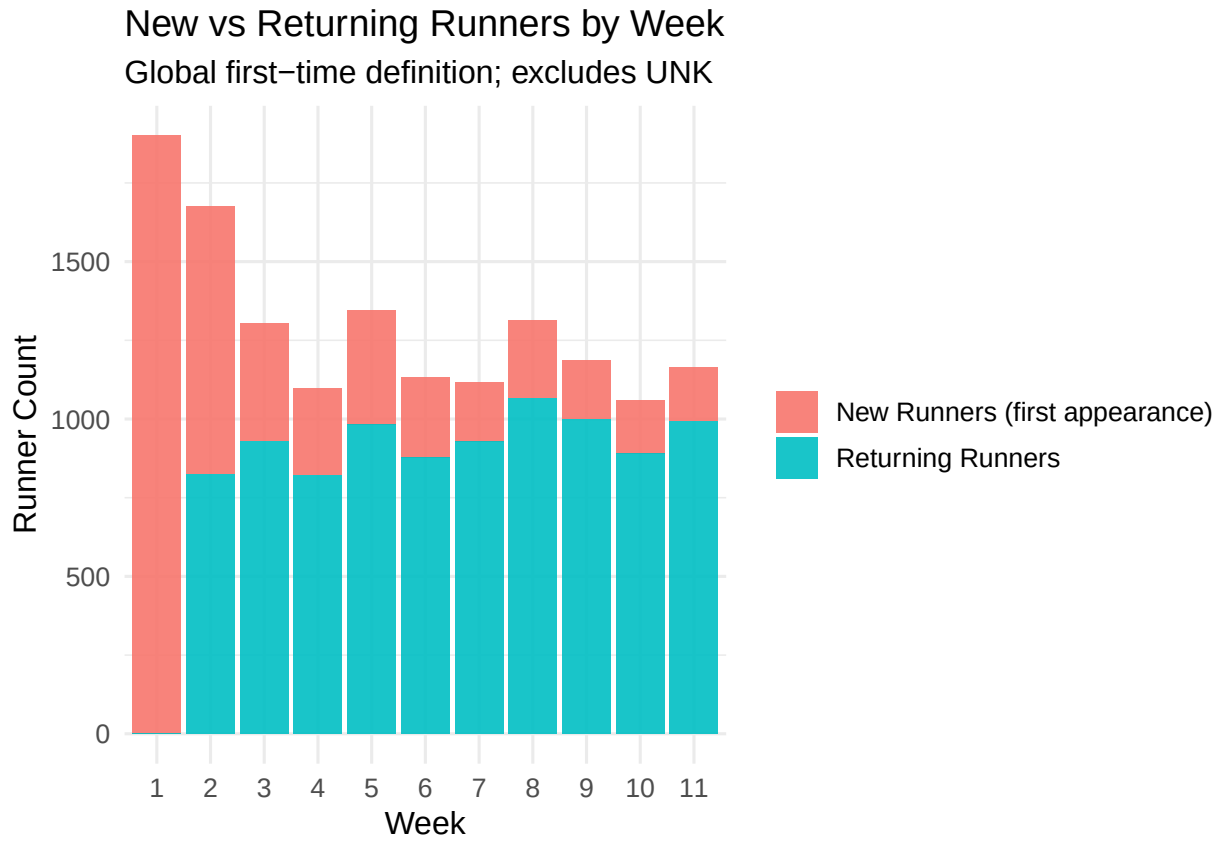
| Week | New Runners | Returning Runners | Total Runners | Percentage of New Runners |
|---|---|---|---|---|
| 1 | 1900 | 0 | 1900 | 100.00 |
| 2 | 851 | 824 | 1675 | 50.81 |
| 3 | 373 | 930 | 1303 | 28.63 |
| 4 | 275 | 822 | 1097 | 25.07 |
| 5 | 364 | 982 | 1346 | 27.04 |
| 6 | 257 | 877 | 1134 | 22.66 |
| 7 | 190 | 928 | 1118 | 16.99 |
| 8 | 245 | 1067 | 1312 | 18.67 |
| 9 | 187 | 1000 | 1187 | 15.75 |
| 10 | 169 | 891 | 1060 | 15.94 |
| 11 | 170 | 994 | 1164 | 14.60 |

```
## Detected First-timer column: First_timer
```

```
## [1] "Q2 (event-level): Weekly first-timers based on data flag"
```

Table 3: Weekly new vs returning runners across every Parkrun

| Week | First Timers | Total Runners | Percentage of First Timers |
|---|---|---|---|
| 1 | 2088 | 1900 | 109.89 |
| 2 | 1682 | 1675 | 100.42 |
| 3 | 1310 | 1303 | 100.54 |
| 4 | 1104 | 1097 | 100.64 |
| 5 | 1353 | 1346 | 100.52 |
| 6 | 1136 | 1134 | 100.18 |
| 7 | 1122 | 1118 | 100.36 |
| 8 | 1324 | 1312 | 100.91 |
| 9 | 1192 | 1187 | 100.42 |
| 10 | 1062 | 1060 | 100.19 |
| 11 | 1170 | 1164 | 100.52 |

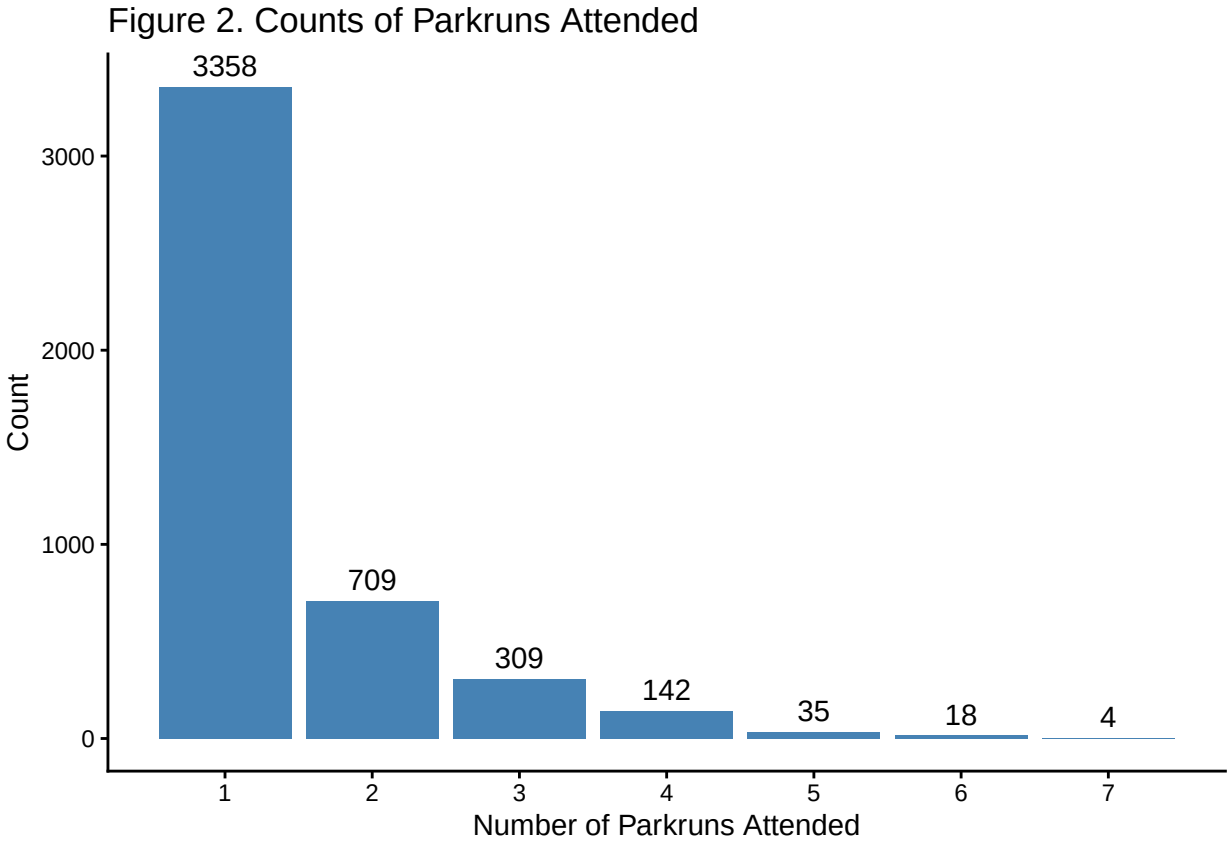


## Question Three: Do people move between Parkruns? How does this affect advertising?

In order to determine whether or not runners attend different Parkruns, we can simply count the number of Parkrun locations each runner has been to in the past 11 weeks. However, due to unidentifiability, we cannot include any unknown runners in this question.

Table 4: Proportion of runners who have attended a different parkrun

| Total Runners | Mixed Runners | Percentage of Mixed Runners |
|---|---|---|
| 4575 | 1217 | 26.6 |




The above table (Table 4) shows that there is a significant number of runners who have attended Parkruns at mulitple locations (26.6%), or about 1200 runners out of 4600. However, Figure 2 shows that the vast majority of runners have only attended events at one Parkrun location, with very few reaching out to others.

However, this 26% overlap means that advertising across all locations is not going to be particularly budget effective, as advertisements will end up being displayed to people who have already bought Puma products. Therefore, advertising across fewer Parkrun locations can be a more economical option, as it would allow for the same effective advertisement reach, at a lower cost when compared to advertising at all locations.

## Question Four: If Puma want to target elite runners, where should they focus their efforts?

In order to target elite runner for the marketing campaign, we can look at the "Age grade" of each runners.

"Age grade" is a percentage that compares runner's time to the world record for runner's age and gender.

This percentage could be understood as: - <50% = beginner - 50-60% = average - 60-70% = good - 70-80% = very good - 80%+ = excellent

A person who has an age grade above 70 would be consider as an elite runner, this applies to both men and women. Here is the table shows the number of elite runners between parkruns.

```
# Overall elite analysis by location
elite_analysis_overall <- clean_data %>%
  group_by(parkrun) %>%
  summarise(
    total_runners = n(),
    elite_runners = sum(elite_runner),
    elite_percentage = elite_runners / total_runners * 100,
    avg_grade = mean(grade, na.rm = TRUE)
  ) %>%
  arrange(desc(elite_runners))%>%
  rename(
    "Parkrun Location" = parkrun,
    "Total Runners" = total_runners,
    "Elite runners" = elite_runners,
    "Elite runners percentage" =  elite_percentage,
    "Average grade" = avg_grade
  )

elite_analysis_overall %>%
  kable(
    caption = "Table 4.1. Overal Elite Analysis by Parkrun Location",
    digits = 2
  )
```

Table 5: Table 4.1. Overal Elite Analysis by Parkrun Location

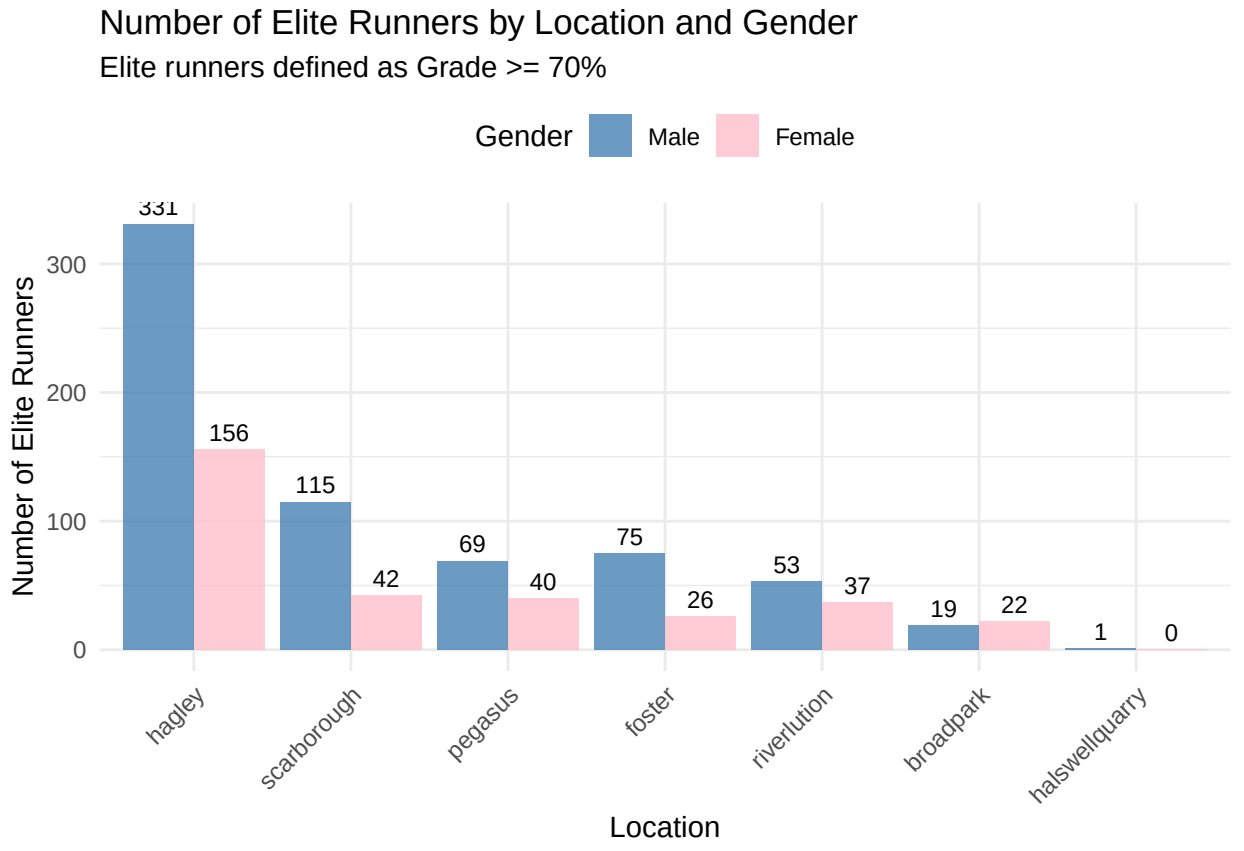
| Parkrun Location | Total Runners | Elite runners | Elite runners percentage | Average grade |
|---|---|---|---|---|
| hagley | 5274 | 487 | 9.23 | 53.02 |
| scarborough | 1669 | 157 | 9.41 | 53.71 |
| pegasus | 1500 | 109 | 7.27 | 52.24 |
| foster | 2018 | 101 | 5.00 | 50.45 |
| riverlution | 1904 | 90 | 4.73 | 49.88 |
| broadpark | 1225 | 41 | 3.35 | 48.81 |
| halswellquarry | 769 | 1 | 0.13 | 43.18 |

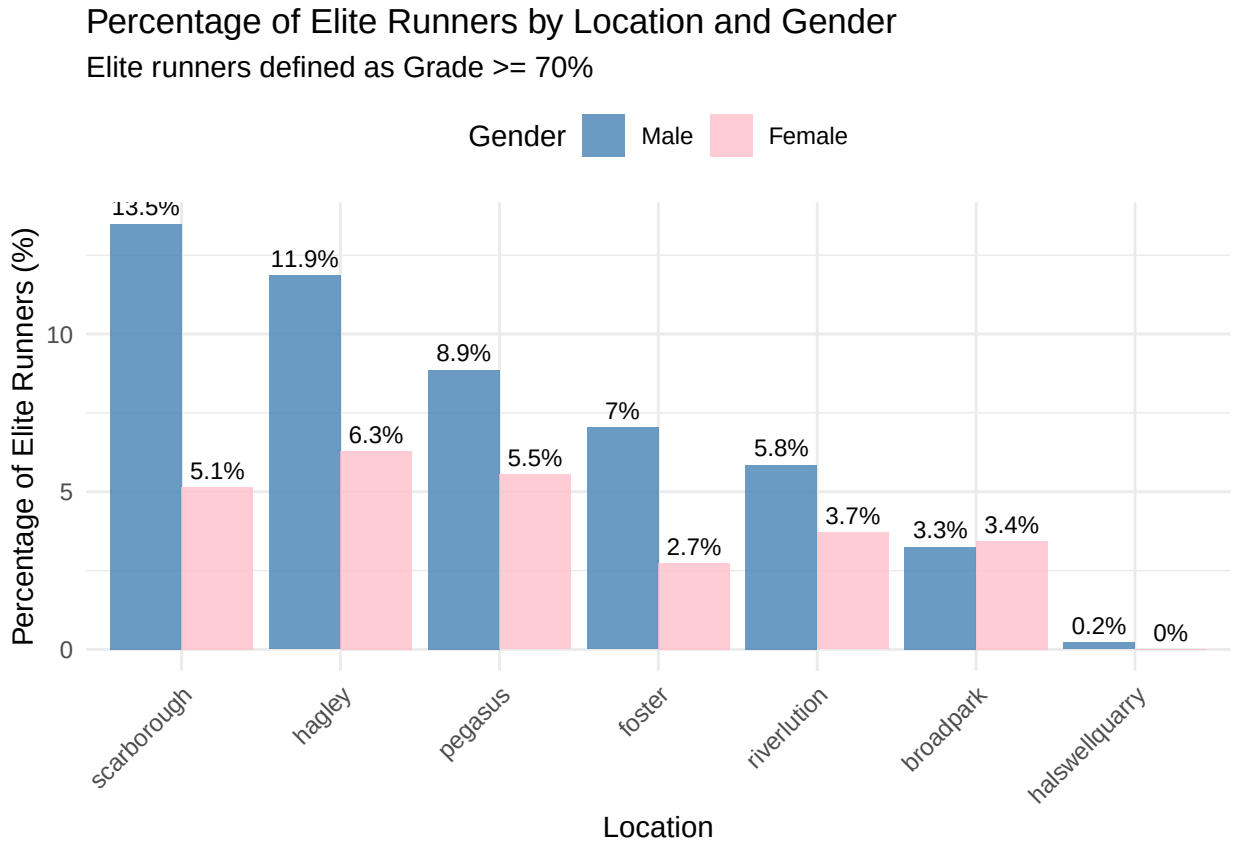
It's noticeable to find that the Hagley parkrun has an outstanding numbers of elite runner but Scarborough parkrun has the highest elite runner percentage among these parkruns.

Table 6: Table 4.2 Elite Analysis by Gender and Parkrun Location

| Parkrun Location | Gender | Total Runners | Elite runners | Elite runners percentage | Average grade | Max grade |
|---|---|---|---|---|---|---|
| broadpark | M | 584 | 19 | 3.25 | 49.74 | 80.28 |
| broadpark | F | 641 | 22 | 3.43 | 47.96 | 84.31 |

| Parkrun Location | Gender | Total Runners | Elite runners | Elite runners percentage | Average grade | Max grade |
|---|---|---|---|---|---|---|
| foster | M | 1066 | 75 | 7.04 | 53.86 | 88.35 |
| foster | F | 952 | 26 | 2.73 | 46.63 | 86.23 |
| hagley | M | 2792 | 331 | 11.86 | 54.97 | 90.27 |
| hagley | F | 2482 | 156 | 6.29 | 50.83 | 92.03 |
| halswellquarry | M | 457 | 1 | 0.22 | 45.33 | 72.54 |
| halswellquarry | F | 312 | 0 | 0.00 | 40.02 | 63.66 |
| pegasus | M | 779 | 69 | 8.86 | 53.51 | 81.87 |
| pegasus | F | 721 | 40 | 5.55 | 50.88 | 85.84 |
| riverlution | M | 906 | 53 | 5.85 | 51.72 | 84.59 |
| riverlution | F | 998 | 37 | 3.71 | 48.22 | 85.74 |
| scarborough | M | 852 | 115 | 13.50 | 56.14 | 85.51 |
| scarborough | F | 817 | 42 | 5.14 | 51.18 | 85.58 |



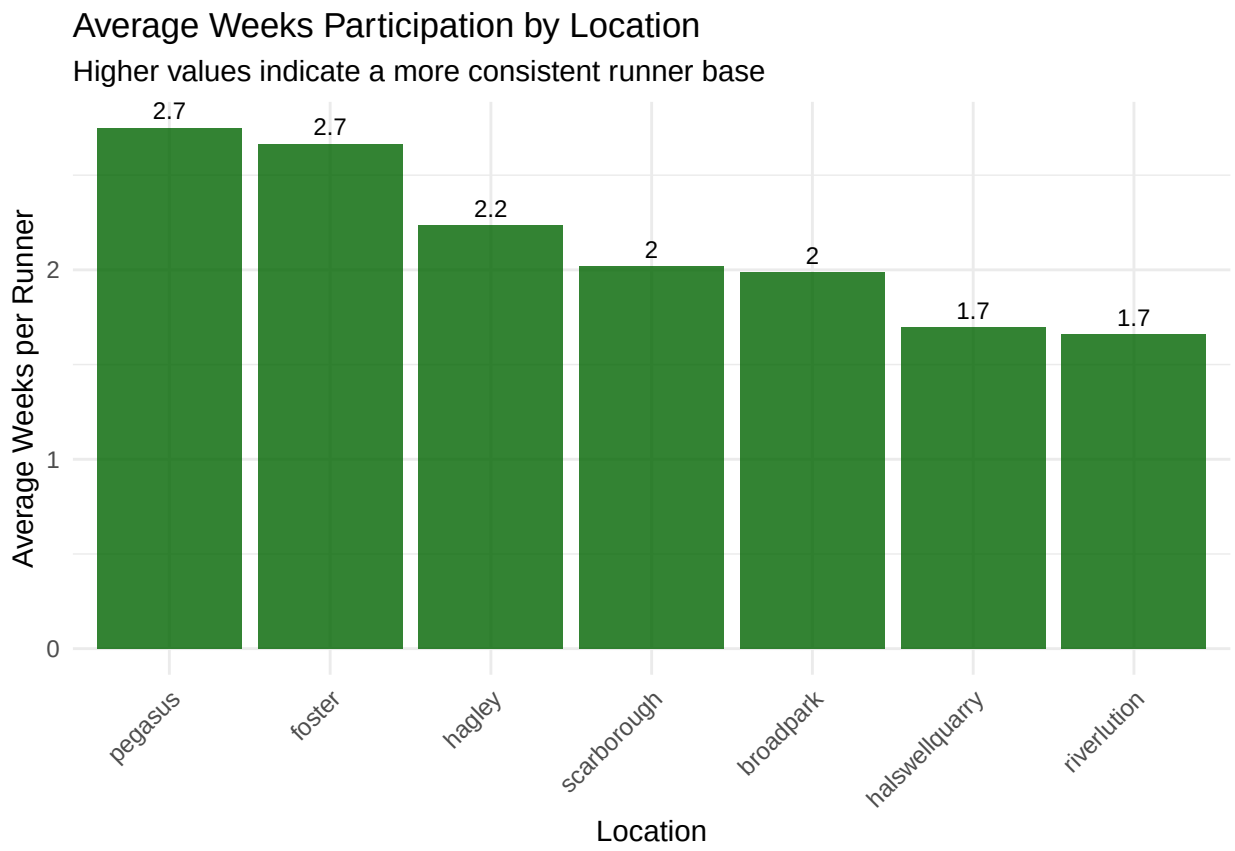


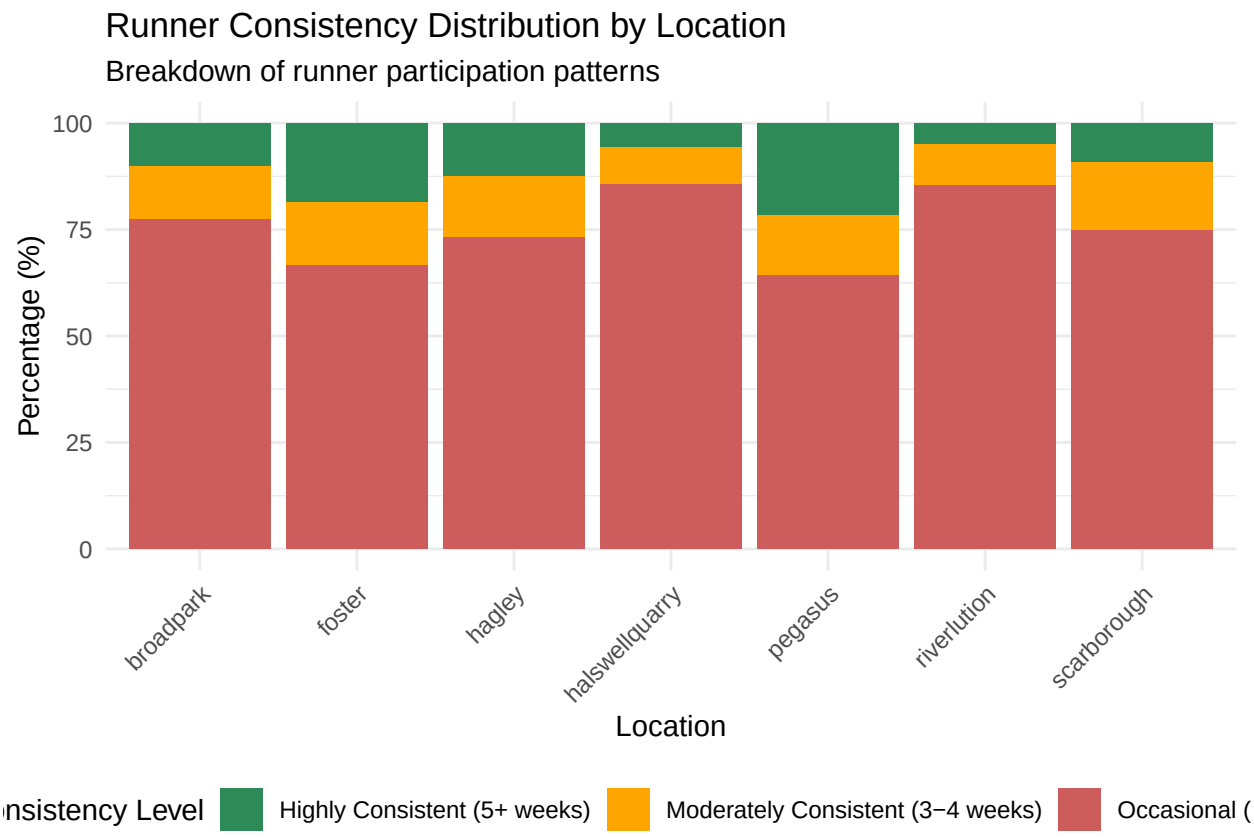

## Question Five: Puma believe more consistent runners have better potential as customers, how many runners are consistent and is consistency different across locations?

table consistent

Table 7: Table 4.1 Number of consistent runner in each parkrun

| Parkrun Location | Total Unique Runners | Highly Consistent (5+ Weeks) | Moderately Consistent (3–4 Weeks) | Occasional (1–2 Weeks) | Percentage Highly Consistent (%) | Average Weeks per Runner |
|---|---|---|---|---|---|---|
| pegasus | 539 | 116 | 76 | 347 | 21.52 | 2.75 |
| foster | 745 | 138 | 111 | 496 | 18.52 | 2.66 |
| hagley | 2300 | 285 | 330 | 1685 | 12.39 | 2.24 |
| scarborough | 809 | 75 | 128 | 606 | 9.27 | 2.02 |
| broadpark | 603 | 61 | 75 | 467 | 10.12 | 1.99 |
| halswellquarry | 449 | 25 | 39 | 385 | 5.57 | 1.70 |
| riverlution | 1131 | 56 | 110 | 965 | 4.95 | 1.66 |


Average Weeks Participation by Location
Higher values indicate a more consistent runner base
Average Weeks per Runner
0
1
2
2.7
2.7
2.2
2
2
1.7
1.7
pegasus
foster
hagley
scarborough
broadpark
halswellquarry
riverlution
Location



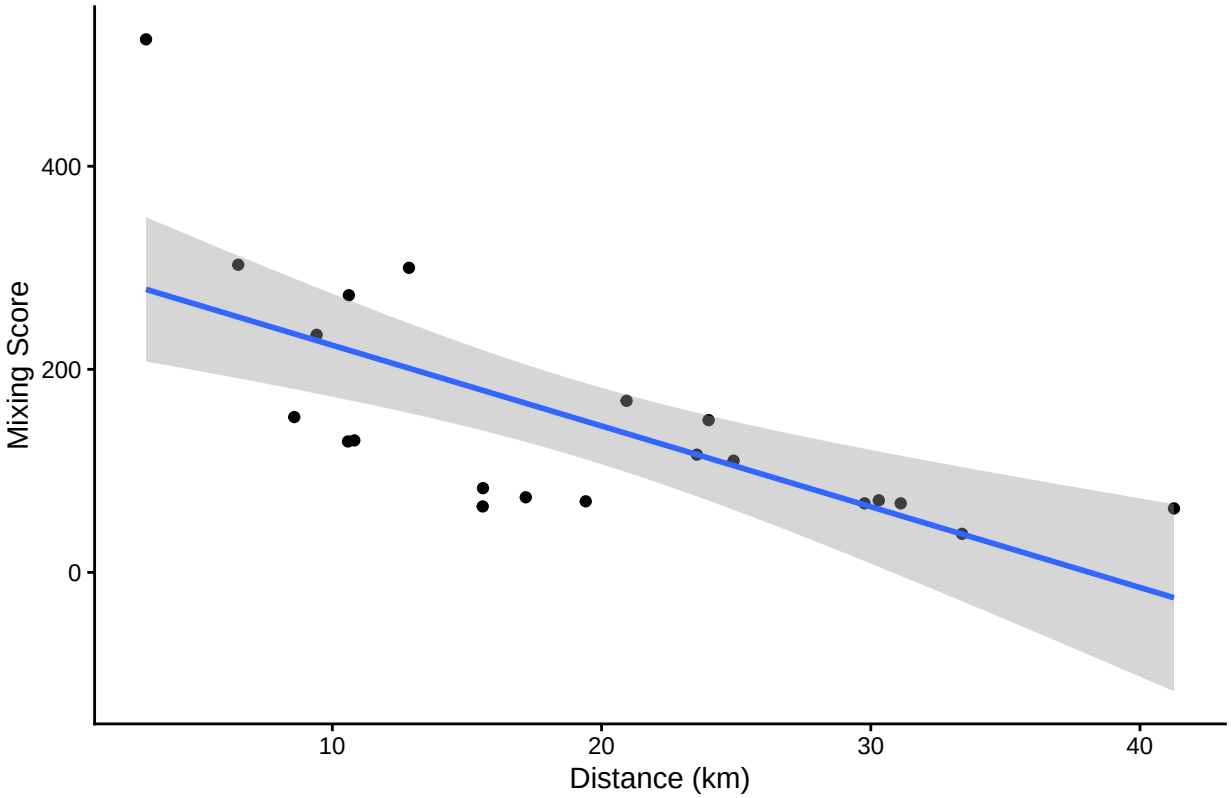

### Question Six: Does distance between Parkruns affect how much mixing occurs?

In order to determine whether or not the distance between two Parkruns predicts how much mixing occurs, we can obtain each unique pair of Parkruns, and then using geographical data, determine the distances between these pairs, effectively creating a map of each Parkrun location.

We can then count the number of runners who have gone to both locations in these pairs, and set that as our Mixing Score. A higher Mixing Score means more runners have attended both Parkruns in the past 11 weeks.

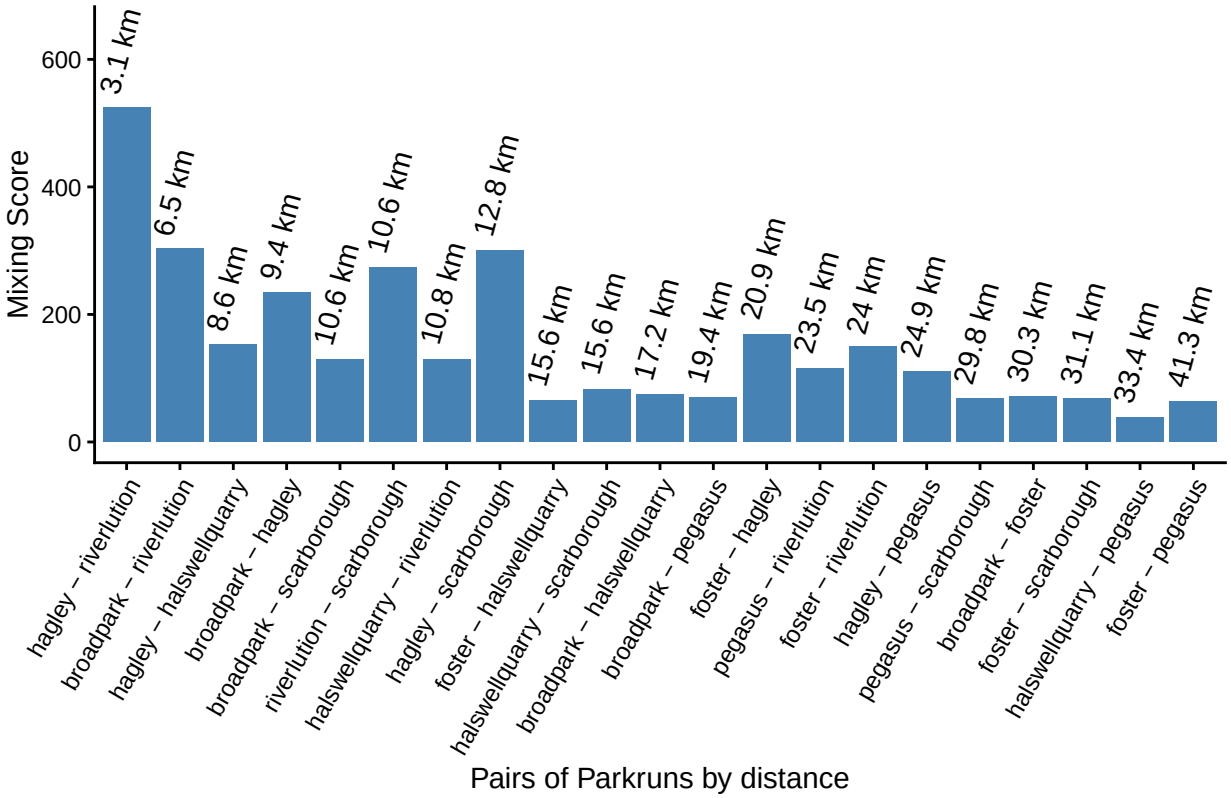
```
## `geom_smooth()` using formula = 'y ~ x'
```

Figure 3. Mixing Score vs. Distance




As seen above, Figure 3 shows that there is a negative correlation between Distance and Mixing Score. In other words, as the distance between parkruns increases, the amount of mixing decreases.

Figure 4. Parkrun Pairs and Mixing Scores arranged by distance




The above bar chart (Figure 4), also shows this correlation, but also labels each parkrun pair with the distance between them. As is seen in the chart, the Hagley and Riverlution parkruns are very close (only 3km apart), and as such, receive a very large amount of mixing.

As for how this affects advertising, this allows for more control over audience overlap. Given the amount of mixing between the Hagley and Riverlution parkruns, advertising at just one of these parkruns would minimise the amount of overlap and spending, without reducing the effectiveness of the advertisements. However, given the amount of distance between the Foster and Pegasus parkruns, advertising at only one location would only harm any marketing efforts.

As such, for pairs of parkruns with high amounts of mixing, advertising at just one location would be the best option, preferably the parkrun which is generally more popular/has the most runners.

## Summary and Conclusion